

Economics Task 6 — Balance of Payments & Terms of Trade

Illustrated study notes. Everything in the condensed version, expanded, with seven diagrams. Charts are built from live RBA data; the theory diagrams are drawn to exact geometry.

Assessed: short answer test, MCQ + short answer/data. Five dot points — the concept and structure of Australia's BOP · reasons for Australia's current account balance · the CAB and the savings/investment gap · the concepts of the terms of trade and the TOT index · the effects of changes in Australia's terms of trade.

1. Definitions — write these verbatim

Balance of payments: a systematic record of all transactions between the residents of Australia and the residents of the rest of the world over a given period of time. *Residents* = individuals, firms, the government and other organisations operating in Australia. It is a **flow** over a period, not a stock.

Current account (CAB): records the value of the flow of **goods, services and income** between Australian residents and the rest of the world. "Current" because what is traded is consumed or received in the current period.

Capital and financial account (KAFA): records capital and financial transactions, including **changes of ownership** of Australia's assets and liabilities.

Terms of trade: an **index** measuring the **relative movements in the prices of exports and imports** — the ratio of export prices to import prices. It measures **the quantity of imports obtainable in exchange for a given volume of exports**.

Savings-investment gap: the gap between Australia's intended level of investment and the level of domestic savings available to fund it.

Servicing costs: future financial obligations arising from borrowing funds overseas (**interest**) or attracting foreign investment (**dividends and profits**).

2. Formulas

CREDIT = money INTO Australia DEBIT = money OUT of Australia

Net goods = goods credits – goods debits

Net services = services credits – services debits

BOGS (trade balance) = net goods + net services

CAB = BOGS + net primary income + net secondary income

KAFA = capital account balance + financial account balance

CAB + KAFA = 0 ← always, by double-entry book-keeping

CAB ≈ national savings (S) – investment (I)

TOT index = $(XPI \div MPI) \times 100$

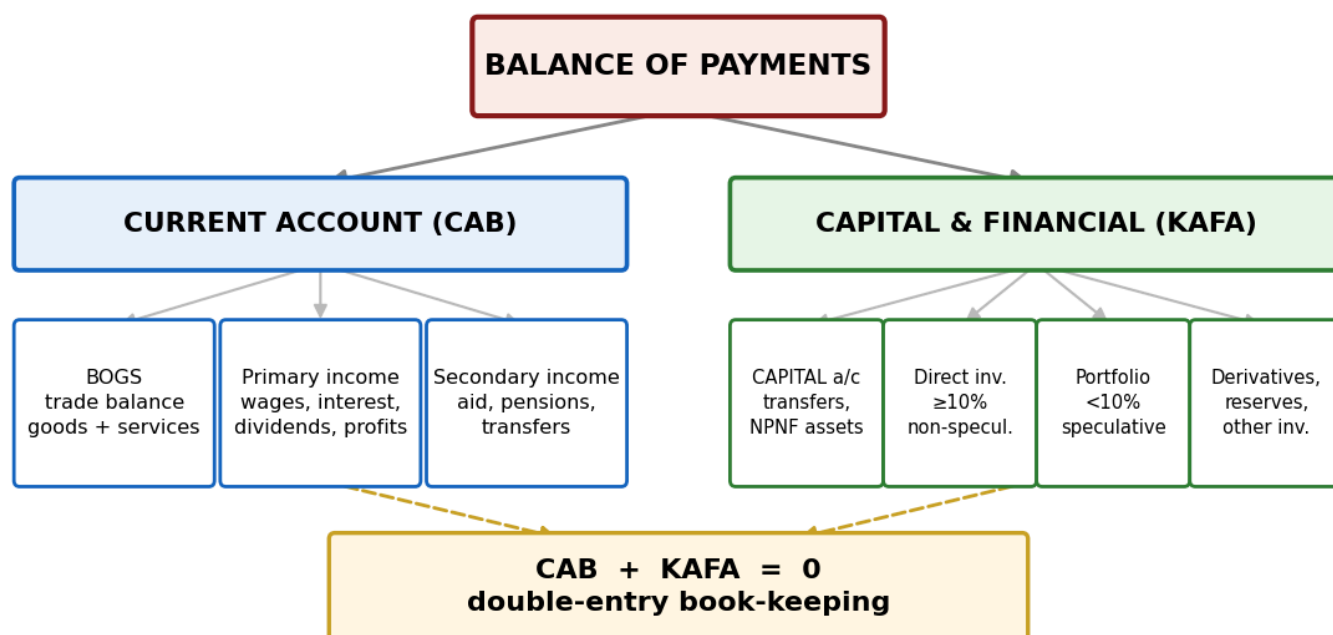
Price index = $(\text{total price this year} \div \text{total price in base year}) \times 100$

% change = $(\text{new} - \text{old}) \div \text{old} \times 100$ ← divide by the OLD value

Net errors and omissions is the balancing item, included because some transactions are hard to measure accurately and others are missed or omitted entirely.

3. Structure of the BOP

Fig 7 Structure of the balance of payments



Account	Component	Records	Examples
Current	BOGS / trade balance	Goods and services exported less imported. Forms part of GDP , which is why most attention goes here	Iron ore, coal, LNG, beef ↔ cars, fuel, machinery; tourism, education, freight, insurance
	Primary income	Income earned from, less paid to, the rest of the world from working and from financial investments	Wages, dividends, interest, profits
	Secondary income	Government income/transfers, and transfers where one party provides something without receiving anything in return	Foreign aid, pensions, insurance payouts, tax refunds
Capital	Capital transfers	Ownership transferred with nothing specific in return	Debt forgiveness; conditional grants for capital projects (aid to build roads, dams, schools)
	Non-produced, non-financial assets	Intangibles and rights to use natural resources	Brand names, copyrights, trademarks; mining and fishing rights
Financial	Direct investment	Long-term investment in a business with ≥10% voting power	Buying a factory; 25% of a company. Non-speculative
	Portfolio investment	Shares or bonds, <10% , no influence over operations	5% of Facebook. More speculative
	Financial derivatives	Contracts whose value derives from another instrument — these trade risk , not funds	Options on rates, currencies, indexes
	Reserve assets	Reserves held by the RBA for FX intervention and IMF commitments	Gold, foreign currency
	Other investment	The residual	Trade credit; currency and deposits

4. Credits, debits and double entry

Current account — intuitive. Provide something of value → **credit**. Receive something of value → **debit**.

KAFA — counter-intuitive. Here credits and debits reflect **changes in ownership of assets and liabilities**, not the direction money moves.

	KAFA entry
Australia's liabilities increase OR assets decrease	CREDIT (+)

	KAFA entry
Australia's assets increase OR liabilities decrease	DEBIT (–)

Why. Giving up an asset means **supplying a financial asset to the rest of the world** → credit. Acquiring a claim overseas means **acquiring a financial asset from the rest of the world** → debit.

So: Australians buy foreign shares → Australia gains an asset → **debit**. Money leaves an Australian bank account to pay for imports → Australia loses an asset → **credit**.

The method for any double-entry question

1. Find the account for the **item** being traded, and its sign.
2. The offsetting **money** entry goes in the **financial account** (usually *other investment — currency and deposits*, or *trade credit*) with the **opposite sign**.

- **Export** → CR in the CAB + DR in the FA (currency)
- **Import** → DR in the CAB + CR in the FA (currency)

Worked — \$2m of wheat exported. CR \$2m in the current account (goods), because Australia provides something of value. DR \$2m in the financial account: the overseas buyer must obtain AUD, producing an inflow of AUD and an outflow of foreign currency, which is either an increase in Australia's foreign assets or a decrease in its foreign liabilities. Net zero.

Worked — \$5,000 Bali holiday paid from an Australian account. DR \$5,000 in the trade balance as a **services import (tourism)**. CR \$5,000 in the financial account under *other investment — currency and deposits*, because Australians now hold \$5,000 less — a **reduction in Australia's financial assets**.

Worked — \$100m iron ore export, paid on arrival. CR \$100m goods. DR \$100m financial account under *other investment — trade credit***, because payment is only made after the goods are received.

Classification drill — cover the right and work down

Transaction	Account	Sub-section	CR/DR
Coal export to China	Current	Goods	CR
Apple iPhone shipment to Australia	Current	Goods	DR
Mining company purchasing trucks	Current	Goods	DR
Chinese company paying for Australian education	Current	Services	CR
Australian spending money in Bali	Current	Services	DR
Purchasing shipping for iron ore	Current	Services	DR
Overseas company paying interest to an Australian company	Current	Primary income	CR
Australian backpacker earns money in London	Current	Primary income	CR
Businessman paying interest on a loan to an overseas investor	Current	Primary income	DR
US shareholder in Wesfarmers receives a dividend	Current	Primary income	DR
Donation to a charity in Africa	Current	Secondary income	DR
Chinese business buying 25% of an Australian company	Financial	Direct investment	CR
Australian company buying 5% of Facebook	Financial	Portfolio	DR
RBA selling gold overseas	Financial	Reserve assets	CR
Drawing on a UK bank account to buy an Australian house	Financial	Other investment	CR
Buying the rights to the Nando's name in Australia	Capital	NPNF assets	DR

Traps. Capital *goods* (trucks, machinery, excavators) are still **goods** in the current account. Brand names, patents and copyrights go in the **capital** account, not the financial account. **10%** is the direct/portfolio boundary. In the KAFA, a credit is **not** "money in".

5. BOP calculations

Worked. Goods credits 600, goods debits 500, net services –300, net income ?, KAFA +1150.

CAB: CAB + KAFA = 0 → CAB = -1150 (you need nothing else)
 Net income: net goods = 600 - 500 = +100
 CAB = 100 + (-300) + net income = -200 + net income
 -200 + net income + 1150 = 0 → net income = -950

Worked — a full table. Export \$40,000 beef; \$6,000 Fiji holiday; a Canadian buys \$15,000 (3%) of an Australian company; an Australian firm buys a \$25,000 German patent.

	Credit	Debit	Total
Current account	+40,000 (beef)	-6,000 (Fiji holiday)	+34,000
Capital and financial	+15,000 (shares) +6,000 +25,000 (currency)	-25,000 (patent) -40,000 -15,000 (currency)	-34,000
Totals	+86,000	-86,000	0

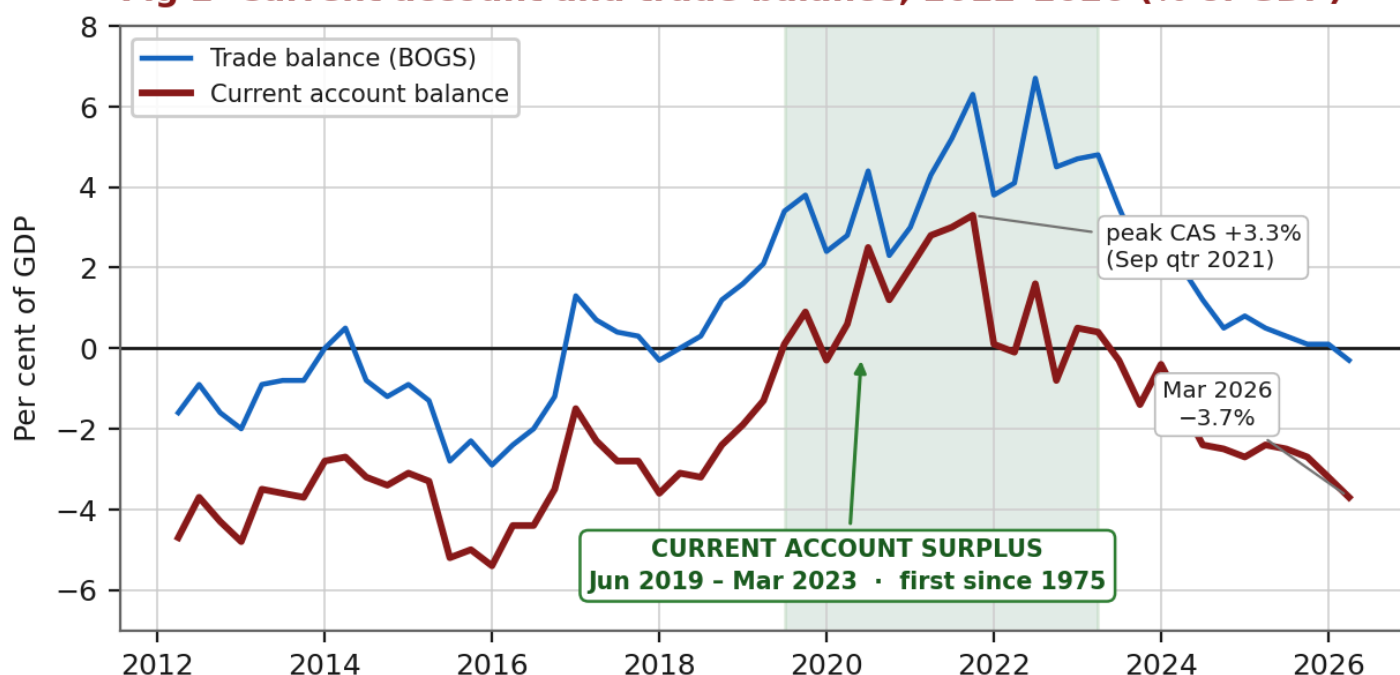
3% < 10% → **portfolio**. A patent → **capital** account, NPNF assets. Every transaction needs **two** entries, and the table must sum to zero.

6. Reasons for Australia's current account balance

CAB = BOGS + NPY + NSY. BOGS is volatile and swings between surplus and deficit; **NPY is persistently in deficit. Often the NPY deficit exceeds the BOGS surplus, producing a CAD.**

Always address **both** components and state **which dominated**. That final step is where the marks are.

Fig 2 Current account and trade balance, 2012-2026 (% of GDP)



Source: RBA Statistical Table I1. The vertical gap between the two lines is the net income deficit — always negative.

Read the chart this way. The blue line is the trade balance, the dark red line is the current account. The **vertical gap between them is the net income deficit** — notice it is always negative, and roughly 2-4% of GDP throughout. The current account only reaches surplus when the trade balance climbs high enough to outweigh that gap, which is exactly what happened from mid-2019.

CAS (surplus)	CAD (deficit)
Net creditor to the rest of the world; providing an abundance of resources and owed money	Net debtor; investing more than it is saving , using other economies' resources for domestic consumption and investment

A deficit is **not necessarily bad**, especially for an economy developing or under reform — sometimes an economy must spend now to earn later, so it runs a deficit intentionally.

Structural vs cyclical — questions ask for this by name

STRUCTURAL	CYCLICAL
Fundamental factors, mainly affecting the income balance	Temporary factors from the business cycle, mainly affecting the trade balance
Savings–investment gap · foreign investment · foreign liabilities	Domestic business cycle · world business cycle · exchange rates · commodity prices · terms of trade
Why the CAD persists	Why the CAD fluctuates

Factors affecting BOGS

Factor	Change	Chain	CAB
Trading partner growth	↑	Exports ↑, particularly minerals and fuels and services → credits ↑ → BOGS ↑	↑
Domestic growth	↑	National income ↑ → households spend more on discretionary and luxury goods → consumption imports ↑; firms raise output → capital and intermediate imports ↑ → debits ↑ → BOGS ↓	↓
Exchange rate	AUD depreciates	Exports cheaper for foreigners, imports dearer → assuming demand for X and M is price elastic → X ↑, M ↓ → credits ↑, debits ↓	↑
Terms of trade	↑	Export prices ↑ relative to import prices → foreigners pay more for our exports → credits ↑	↑
Competitiveness	↑	Via relative wages, the inflation differential and exchange rates → exports more attractive → credits ↑	↑
External shocks	—	COVID: X and M both fell, but M fell by more → BOGS ↑	either

Two things markers look for: **domestic growth moves the CAB the opposite way**, and the exchange rate chain **requires the elasticity assumption** to be stated.

Factors affecting net primary income

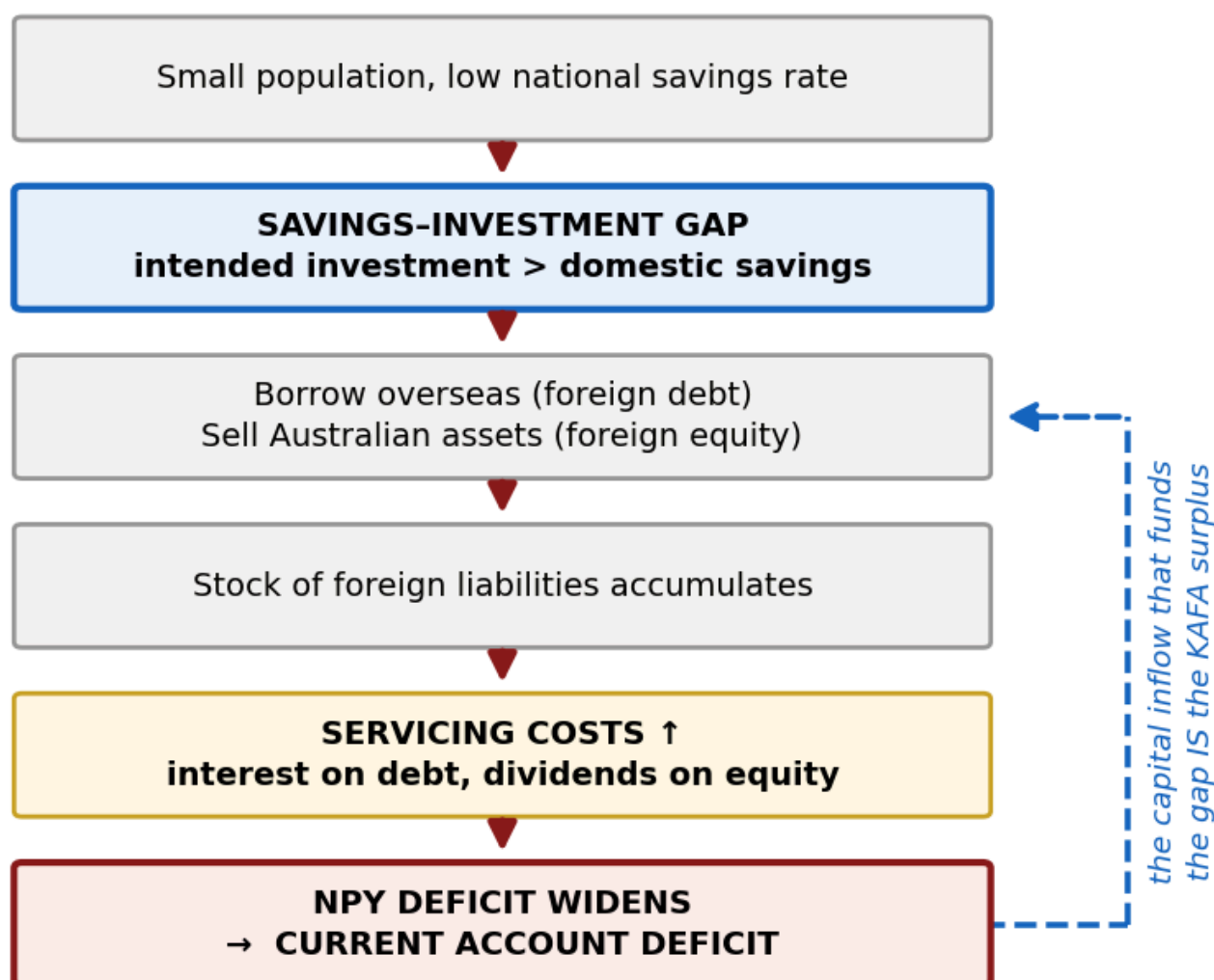
NPY = servicing costs. Foreign **debt** → **interest**. Foreign **equity** → **dividends and profits**. The NPY is persistently in deficit because Australia is a **net capital importer**, so payments to foreign investors exceed the income we receive on our investments abroad.

The deficit widens with: a **bigger S-I gap**; a **larger accumulated stock** of foreign liabilities; **higher interest rates**; and **higher profits of foreign-owned Australian firms**.

The interaction worth a mark. A rising TOT lifts mining profits, but much of Australian mining is **foreign-owned**, so a large share is paid out as dividends overseas — a primary income debit. A TOT boom therefore improves BOGS while **worsening** the NPY deficit, partly offsetting the gain in the CAB.

7. The savings-investment gap

Fig 5 The savings-investment gap and the CAD



Reverse every arrow to answer “what if national savings rose?”

Australia has a **small population** and a **relatively low national savings rate**, creating a **gap between intended investment and available domestic savings**. To close it, Australia **borrow from overseas (foreign debt)** and **sells ownership of Australian assets (foreign equity)**. Both create **servicing obligations** — interest on debt, dividends and profits on equity — recorded as **primary income debits**. This **increases NPY outflows, widens the NPY deficit and decreases the CAB**. The capital inflow that funds the gap **is the KAFA surplus**, the mirror image of the CAD.

The identity: $CAB \approx S - I$. If $I > S$ the gap is funded by foreign capital → **CAD** plus a matching KAFA surplus. If $S > I$, Australia is a net lender → **CAS** plus a KAFA deficit.

Why the gap exists: small population → small savings pool; low savings rate; **high investment requirements** as a capital-intensive resource economy (mines, LNG plants, rail, ports, and lately data centres and the energy transition); and Australia is an attractive destination for foreign capital.

Reverse it — a guaranteed question. Higher national savings → **smaller S-I gap** → firms fund projects from **domestic savings rather than foreign capital** → **less foreign borrowing and equity** → **fewer NPY outflows** → **smaller NPY deficit** → **CAB rises, CAD shrinks**. And, consistently with the identity, a **smaller KAFA surplus**.

Is a CAD a problem? Not if foreign capital funds **productive** investment returning more than its servicing cost, and if the borrowing is voluntary private-sector activity — the *consenting adults* or **Pitchford** view. It is a problem if it funds **consumption** rather than productive investment, or if rising servicing costs and dependence on capital inflow leave Australia exposed to a **sudden stop** or higher global rates. **Conclusion: it depends on what the capital is used for, not on the size of the deficit.**

8. The terms of trade and the TOT index

Rise = FAVOURABLE. XPI up relative to MPI → we buy **more** imports per unit of exports → **purchasing power up**.
Fall = UNFAVOURABLE. MPI up relative to XPI → **fewer** imports per unit of exports → purchasing power down.

The index level is meaningless on its own — only the change matters.

Why it matters: changes in the XPI and MPI affect the value of total exports and imports, and therefore the **current account**, the **exchange rate** and **national income**.

Both indexes are **weighted** averages, so a large price move in a heavily weighted item drives the whole index — **iron ore** in the XPI, **oil** in the MPI.

Building an index from raw data

Total the prices for each year; assign a **base year = 100**; every other year is $(\text{total} \div \text{base total}) \times 100$.

	Y1	Y2	Y3
Total export prices	\$420	\$410	\$470
XPI	100 (base)	97.6	111.9

Worked calculation

XPI 103, MPI 117 → $(103/117) \times 100 = 88.0$
 XPI 109, MPI 119 → $(109/119) \times 100 = 91.6$
 % change = $(91.6 - 88.0) \div 88.0 \times 100 = +4.1\%$ → favourable

The traps

Percentage points ≠ per cent. 95.5 → 97.4 is a rise of **1.9 index points** but **2.0 per cent** ($1.9 \div 95.5$).
 Read the wording, and always divide by the **older** value.
Price indexes tell you nothing about values or volumes. Given only XPI and MPI you cannot conclude anything about export revenue or quantities. This is the most common MCQ distractor in the topic.
"TOT fell" does not mean "exports fell". The TOT can fall while export prices *rise*, if import prices rise by more.

The four combinations

TOT rises when: XPI ↑ and MPI flat · MPI ↓ and XPI flat · **both ↑ but XPI ↑ by more** · **both ↓ but MPI ↓ by more**. Flip all four for a fall.

MCQ shortcut: $\% \Delta \text{TOT} \approx \% \Delta \text{XPI} - \% \Delta \text{MPI}$. For June quarter 2026: XPI +1.1%, MPI +5.7% → $\approx -4.6\%$ (exact -4.35%). Close enough to pick the option; use the full formula if asked to *calculate*.

9. Factors affecting the terms of trade

Australia is a **price taker** for both exports and imports, because we are a medium-sized economy — **world markets set our X and M prices, not us**.

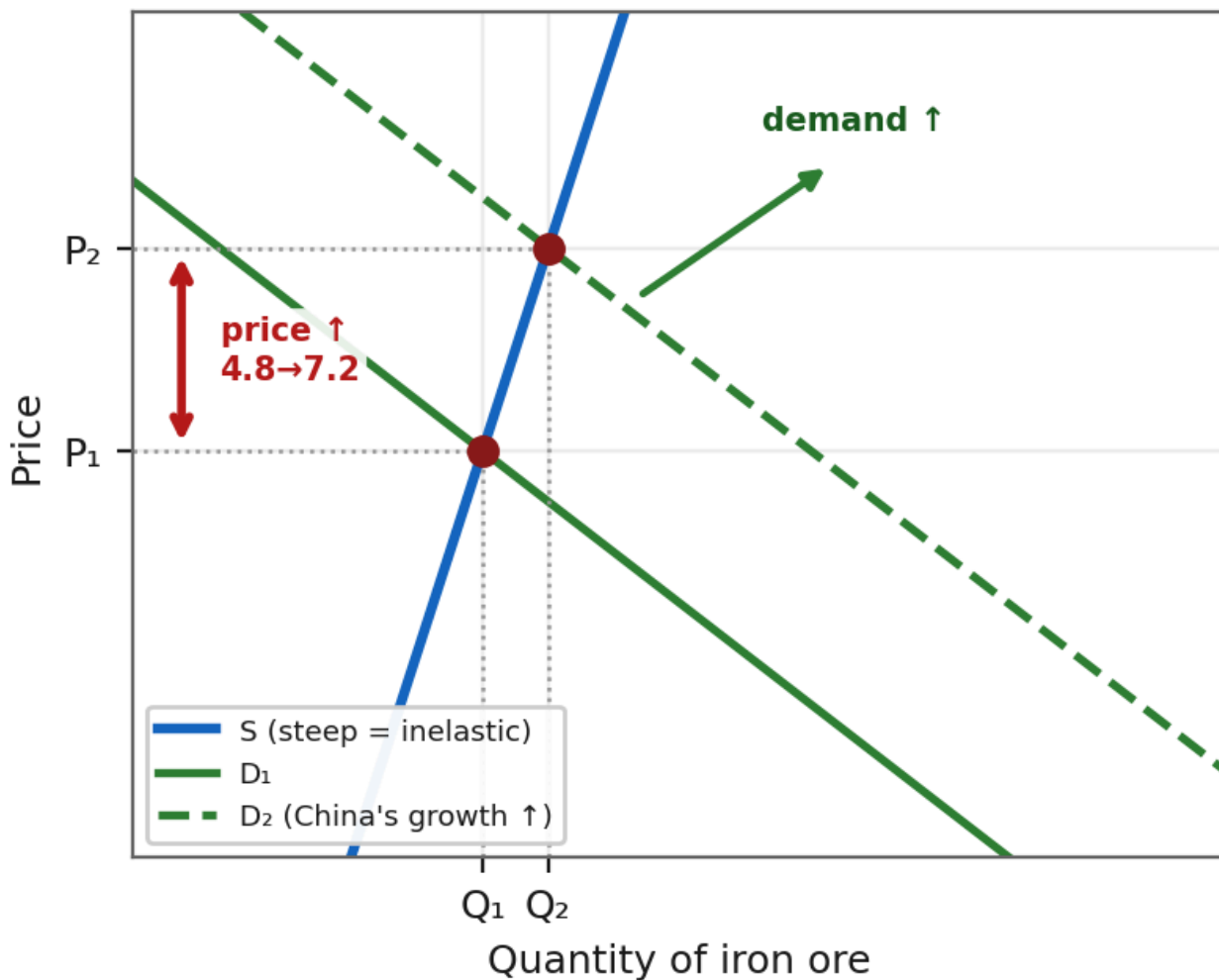
Commodity prices — the dominant driver

About **75%** of Australia's exports are commodities. Their **demand and supply are both inelastic**, giving **steep curves**, so **small shifts in D or S produce large price swings**.

Why demand is inelastic	Why supply is inelastic
Few global substitutes	Long production lags — mines and crops take years to adjust
A small proportion of the final good's cost	High capital requirements — hard to scale quickly

Why demand is inelastic	Why supply is inelastic
A necessity for production — can't stop using them short-run	Geographic and climate constraints
Long-term global contracts dampen responsiveness	Regulatory and infrastructure limits

Fig 3 The iron ore market



Small demand shift → large price rise → XPI ↑ → TOT ↑ → BOGS ↑, CAB ↑, national income ↑

When China's economy expands, world demand for iron ore shifts right ($D_1 \rightarrow D_2$). Along a **steep supply curve** the price rises sharply from P_1 to P_2 with only a small quantity increase. Exporters sell **more at a higher price** → XPI ↑ → TOT ↑ → BOGS ↑ → CAB ↑ and **national income** ↑. Reverse it for a Chinese slowdown.

Manufactured goods prices

Most of Australia's **imports** are manufactured and intermediate goods — appliances, equipment, machinery, ICT. Their **supply is elastic** (a flat supply curve), so demand shifts have **little effect on price**. As China became the world's largest manufacturer, and with productivity and technology gains, manufactured and ICT prices have **trended down** → MPI ↓ → TOT ↑.

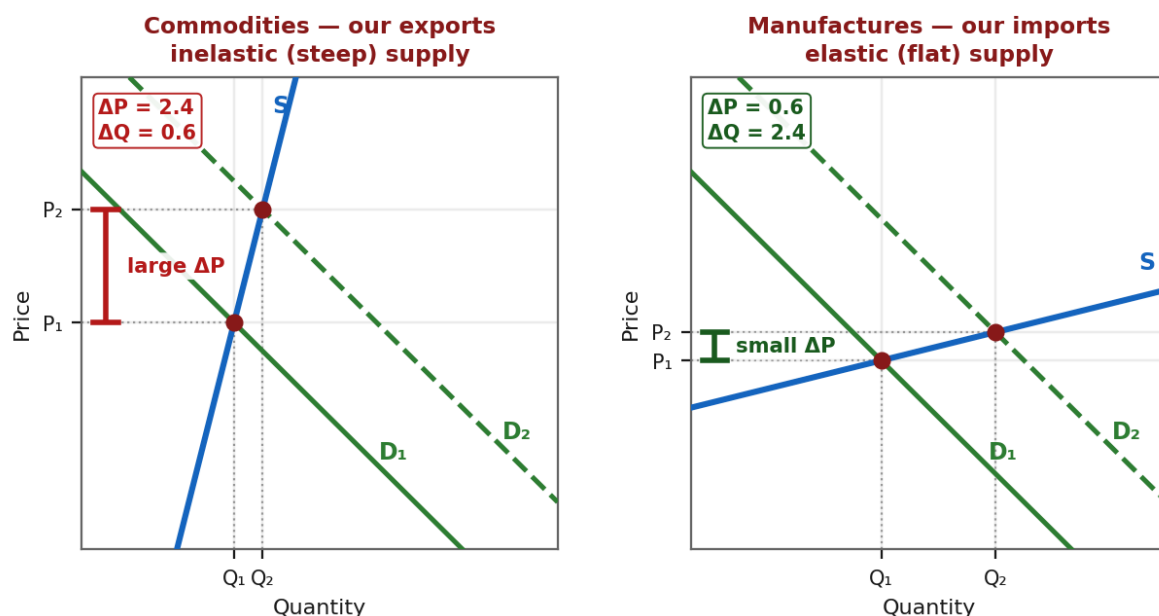
Exception: during COVID, **supply bottlenecks** cut supply → import prices ↑ → MPI ↑ → TOT ↓.

Oil prices

Volatile and inelastic, and Australia is a **significant importer**. Higher oil prices raise **energy and transport costs across nearly all import prices**. But they also lift the price of our energy **export substitutes** — coal and LNG — so the net TOT effect depends on the relative weights. In 2016-19 the export effect dominated via LNG; in 2022 both indexes rose together.

Why the TOT tracks the XPI, not the MPI

Fig 4 Why the terms of trade tracks the XPI, not the MPI

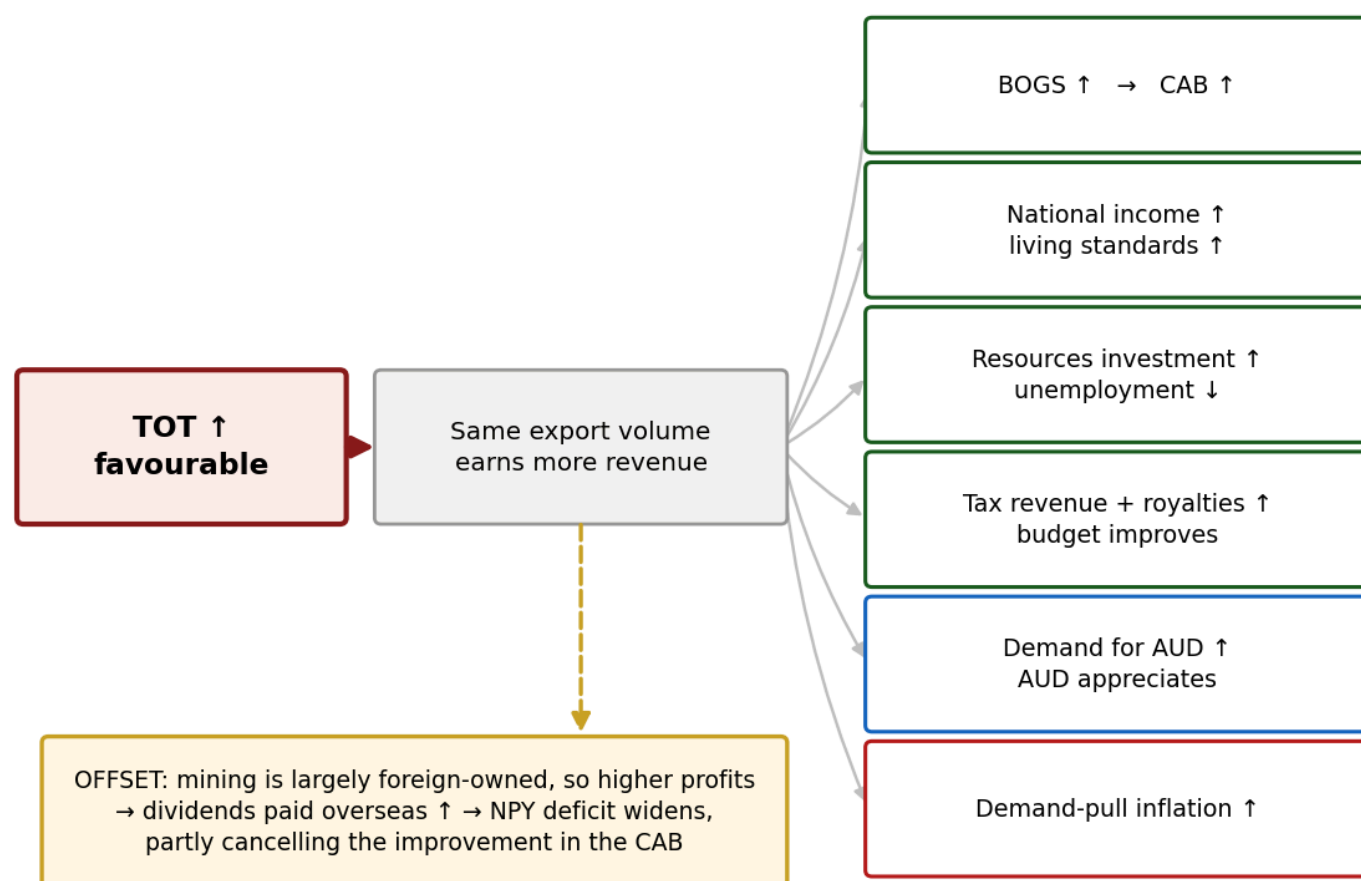


An identical rightward demand shift ($D_1 \rightarrow D_2$) in both markets. Inelastic commodity supply converts it into a large price rise and little extra quantity; elastic manufactures supply absorbs it in quantity with barely any price change. Hence a volatile XPI, a stable MPI, and a TOT driven by export prices.

This is the single most useful diagram in the topic. **The same demand shift** produces a price rise **four times larger** in the commodity market than in the manufactures market, because commodity supply is inelastic and manufactures supply is elastic. Hence a **volatile XPI**, a **stable MPI**, and a TOT driven almost entirely by **export prices**. Over the past decade the XPI ranged over about **60** index points against the MPI's **29**.

10. Effects of changes in the terms of trade

Fig 6 Effects of a rise in the terms of trade



A fall in the terms of trade reverses every one of these.

A rise means the **same volume of exports earns more revenue**. Don't just list the effects — give the mechanism.

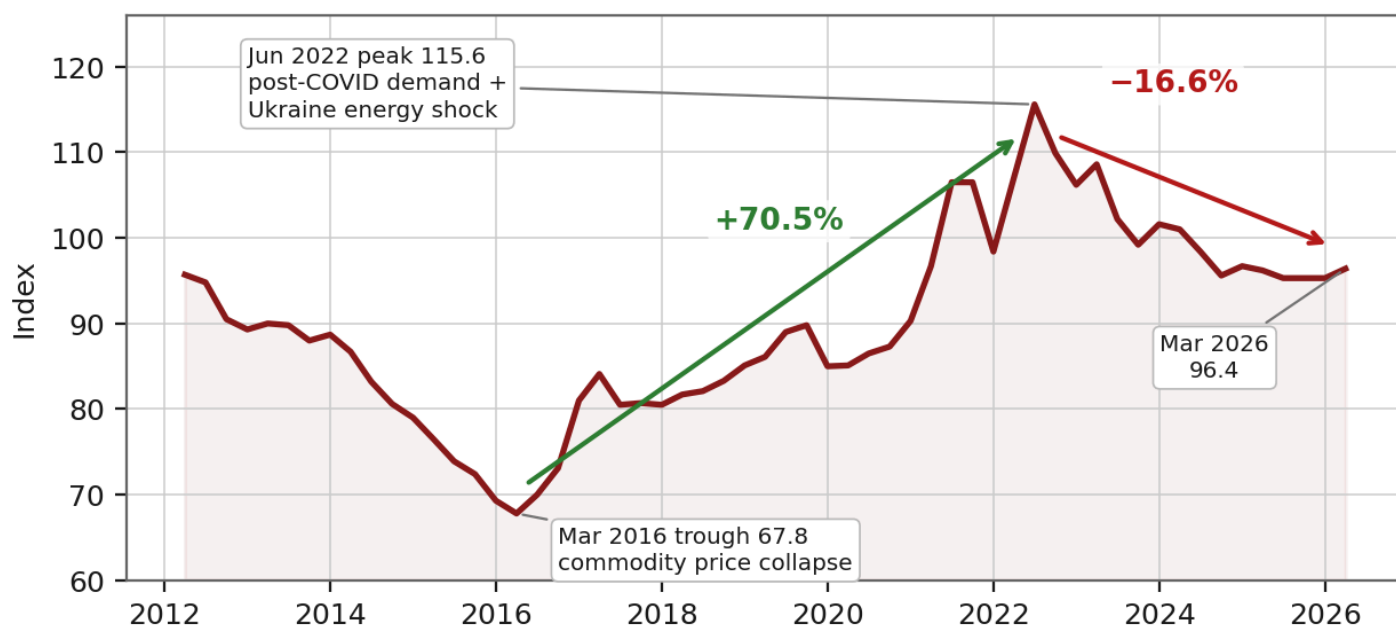
Effect	Mechanism
BOGS ↑ → CAB ↑	Export credits rise relative to import debits
National income ↑, living standards ↑	Higher export revenue becomes higher incomes and profits; more imports per unit of exports raises real purchasing power
Investment ↑, unemployment ↓	Higher expected profitability of resource projects → investment ↑ → derived demand for labour ↑
Government revenue ↑	Company and income tax receipts plus state royalties ↑ → budget improves
AUD appreciates	Foreign buyers must buy more AUD; higher expected returns attract capital inflow → demand for AUD ↑
Inflation ↑	Higher incomes and profits raise aggregate demand → demand-pull pressure, especially near capacity — <i>partly offset</i> because an appreciating AUD makes imports cheaper

Causes of a rise: higher commodity prices; natural disasters lifting rural prices; falling global oil prices; lower manufacturing costs; technology lowering ICT prices. **Causes of a fall:** lower commodity prices; higher global oil prices; supply bottlenecks; higher manufacturing costs.

A fall reverses every row. And note: a falling TOT **reduces** growth prospects and **reduces** national economic welfare — a common MCQ has you pick the opposite.

11. Trends over the last ten years

Fig 1 Australia's terms of trade, 2012-2026 (quarterly)



Source: RBA Statistical Table H1 (ABS national accounts basis). A rise = favourable.

Period	Movement	Why
2014-16	TOT and CAB falling	Commodity prices collapse — iron ore US\$180→US\$40/t (2011-15); China slowing; global oil US\$48→US\$32 in late 2015 as sanctions on Iran ended. MPI stable. TOT troughed at 67.8 in Mar 2016
2016-19	TOT rising , CAB rising but still CAD	XPI up ~40% on coal and iron ore with strong Chinese growth; OPEC stabilised supply, lifting oil and therefore LNG demand. MPI also rose but by less . NPY deficit widened too, as profits of largely foreign-owned firms rose
2020	TOT small fall ; CAB rising	COVID recession cut XPI (LNG, coal); MPI fell too (oil collapse, strong AUD). M fell more than X → BOGS ↑
2020-22	TOT rising to its peak	Recovery demand for raw materials; Ukraine war lifted energy. Peak 115.6 in Jun 2022 — a +70.5% rise from the 2016 trough
Jun 2019 - Mar 2023	CURRENT ACCOUNT SURPLUS — first since 1975	Driven by a record BOGS , peaking at 6.7% of GDP , on bulk commodity prices, with iron ore driving record metal ores exports. NPY stayed in deficit — so the CAS was a trade story, not an income one. Peak CAS +3.3% of GDP in Sep 2021
2023-25	TOT and CAB falling , back to CAD by 2024	China slowdown → iron ore ↓ → XPI ↓ (MPI fell too, but XPI fell by more). Domestic recovery → income ↑ → imports ↑. Rising foreign investment → dividends ↑ → NPY deficit ↑
2025-26	CAD widening sharply	Mar qtr 2026 CAD \$27.1b , up from \$23.0b, and -3.7% of GDP . Primary income paid overseas +\$1.6b on stronger dividends; trade surplus fell \$5.2b to \$3.9b; first monthly goods deficit since Dec 2017 . TOT down -16.6% from its peak
Jun qtr 2026	TOT ≈ -4.4%	XPI +1.1% vs MPI +5.7% — the biggest MPI rise since Dec 2021, driven by fuel. Iron ore weak on Chinese stockpiles, strong global supply and anti-dumping measures

KAFA is the mirror of the CAB. Australia has historically been a **net recipient of capital inflow**, supplementing domestic savings to fund high investment — a **KAFA surplus, the counterpart of our CAD**. In recent years there have been periods of **net capital outflow**, with Australians investing more abroad than foreigners invested here — a **KAFA deficit**, consistent with the shift to a current account surplus. If a question tells you one, immediately state the other.

△ **Index levels differ between series — read the data table you're given.** Your class deck quotes a TOT peak of **128 index points** in the September quarter 2021. The chart above shows a peak of **115.6 in June 2022**. Both can be correct: the class figure is the **goods-only trade price index** on an **older base year**, while this chart is the **national accounts TOT** (goods *and* services) on the current base. Index numbers are **not comparable across different bases or series** — which is exactly why the syllabus stresses that **only the change matters**. In the exam, use whatever indexes the data table gives you and never mix series.

12. Statistics worth quoting

CAD **\$27.1b / -3.7% of GDP** (Mar qtr 2026), up from \$23.0b · first monthly goods deficit since **Dec 2017** · trade surplus fell **\$5.2b to \$3.9b** · primary income paid overseas **+\$1.6b** · **XPI +1.1% vs MPI +5.7%** (Jun qtr 2026) → TOT ≈ **-4.4%** · TOT trough **67.8** (Mar 2016) → peak **115.6** (Jun 2022) = **+70.5%**, then **-16.6%** to **96.4** · CAS **Jun 2019 - Mar 2023**, first since **1975**, peaking at **+3.3% of GDP** · trade balance peaked at **6.7% of GDP** · iron ore **US\$180 → US\$40/t** (2011-15) · commodities **~75%** of exports · XPI range **60** vs MPI range **29**

13. Exam technique

Command words. *Identify/State* — one word or figure, no explanation. *Define* — formal wording. *Calculate* — formula, substitution, answer **with units**. *Describe* — what happened: direction, magnitude, dates. *Distinguish* — both items with an explicit "whereas". *Explain* — why, as a **causal chain**. *Analyse* — components and relationships, using the data. *Evaluate/Discuss* — both sides plus a justified judgement.

Every "explain" answer:

cause → effect on X/M or credits/debits → effect on BOGS or NPY → effect on the CAB

Calculations. Write the formula first — often worth a mark by itself. Substitute visibly. Give **units** (% , \$m, index points). Divide by the **old** value. Check whether the question wants **index points** or **per cent**. State the **direction**, and for the TOT say **favourable** or **unfavourable**.

Trend answers need four things: overall direction · quantified start and end with dates · any turning point · the cause as a chain.

Graph reading. State the units. For "identify the value" questions markers accept a **range**, so read the axis carefully but don't agonise — just never omit the unit.

Before you hand it in. Units on every calculation · an arrow chain in every "explain" · a figure and a date in every trend answer · **both BOGS and NPY** addressed in every CAB question · favourable/unfavourable stated in every TOT question · the **elasticity assumption** stated in any exchange rate chain.

Sources. Charts built from RBA Statistical Tables H1 (terms of trade) and I1 (international trade and balance of payments); quarterly figures from ABS Balance of Payments and International Trade Price Indexes releases. Content summarised and rephrased for licensing compliance.